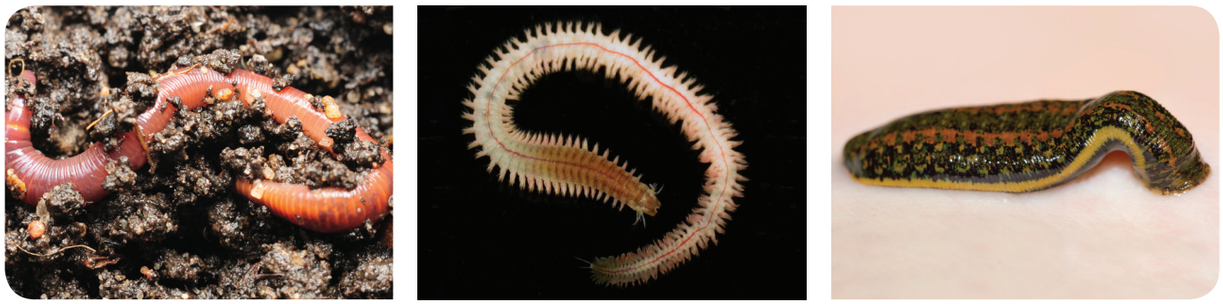


Earthworm *Nereis* *Leech*

Figure 2.14 Examples of Annelida

symmetrical, schizocoelomates and exhibit organ system level of body organisation. The coelom with coelomic fluid creates a hydrostatic skeleton and aids in locomotion. Their elongated body is metamerically segmented and the body surface is divided into segment or metameres. Internally the segments are divided from one another by partitions called septa. This phenomenon is known as **metamerism**. The longitudinal and circular muscles in the body wall help in locomotion. Aquatic annelids like *Nereis* have lateral appendages called **parapodia**, which help in swimming. Chitinous setae in Earthworms, and suckers in Leech help in locomotion. The circulatory system is of closed type and the respiratory pigments are **haemoglobin** and **chlorocruorin**. Nervous system consists of paired ganglion connected by the lateral nerves to the double ventral nerve cord. They reproduce sexually. Development is direct or indirect and includes a **trochophore larva**. Some are monoecious (earthworms) while some are dioecious (Neries and Leech). (Figure 2.14)

Examples: *Lampito mauritii* (earthworm), *Neries* (sand worm), *Hirudinaria* (leech).

How is cephalisation advantageous to animals in finding food?

Filariasis has been a major public health problem in India next only to malaria. The disease was recorded in India as early as 6th century B.C. by the famous Indian physician, Susruta in his book **Susruta Samhita.** In 7 th century A.D., **Madhavakara** described signs and symptoms of the disese in his treatise ' Madhava Nidhana' which holds good even today. In 1709, Clarke identified elephantoid legs in Cochin. The **microfilariae** in the peripheral blood was first identified by Lewis in 1872 in Calcutta (Kolkata).

2.3.7 Phylum: Arthropoda

(G. *arthros*- jointed; podes- feet)

This is the largest phylum of the Kingdom Animalia and includes the largest class called **insecta** (total species ranges from 2-10 million). They are bilaterally symmetrical, segmented, triploblastic and schizocoelomate animals with organ system grade of body organisation. They have jointed appendages which are used for locomotion, feeding and are sensory in function. Body is covered by chitinous exoskeleton for protection and to prevent water loss, It is shed off periodically by a process called **moulting** or **ecdysis**. The body consists of a head, thorax, and abdomen with a body

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